



AICTE-EDUSKILLS VIRTUAL INTERNSHIP • AWS ACADEMY CURRICULUM

Cloud Virtual Internship

An 8-Week Hands-On Journey into AWS Cloud Computing

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IILM University, Greater Noida, U.P.

Duration: April – June 2026

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GRADE

Program Introduction

The rapid adoption of cloud computing created a strong demand for professionals skilled in cloud architecture and deployment. To bridge the gap between academia and industry, AICTE — with the Ministry of Education's National Internship Portal and EduSkills — launched this Virtual Internship Program. Over 8 weeks, the curriculum, provided by AWS Academy, delivered hands-on exposure to AWS fundamentals, core services, and best practices for secure, scalable, cost-efficient cloud solutions.

Delivered By



AICTE

Statutory body under the Ministry of Education overseeing technical education nationwide.



National Internship Portal

Connects students with internships offered by industry and academic partners.



EduSkills

Designs industry-aligned virtual internships — "Nation Building Through Skills."



AWS Academy

Amazon's global education program providing free, ready-to-teach cloud curriculum.

Problem Statement & Objectives



Why This Internship

Traditional CS curricula often lag behind fast-evolving cloud technologies. Students graduate with strong theory but limited exposure to real-world platforms. A structured, hands-on program was needed to build practical experience in provisioning, configuring, and managing cloud resources — preparing students for roles in cloud engineering, DevOps, and infrastructure management.

Scope

AWS Cloud Foundations & Cloud Architecting curriculum, delivered through guided labs, quizzes, and a capstone-style exercise in a sandboxed AWS environment — no production deployment.

Project Objectives

1

Understand core cloud concepts — deployment models and IaaS/PaaS/SaaS.

2

Gain hands-on experience with EC2, S3, IAM, VPC, RDS, and Lambda.

3

Learn cloud architecture principles: scalability, availability, security, cost.

4

Design and deploy a sample cloud-based solution using AWS services.

5

Build industry-aligned skills and earn a recognized certification.

Technologies & Tools Used

Core AWS services studied and used through hands-on labs during the internship.



Amazon EC2

Scalable virtual servers / compute instances.



Amazon S3

Object storage & static content hosting.



AWS IAM

Secure user, role & permission management.



Amazon VPC

Isolated, secure network environments.



Amazon RDS

Managed relational database deployment.



AWS Lambda

Serverless, event-driven computing.



Amazon CloudWatch

Resource & application performance monitoring.



Console & CLI

Provisioning & managing cloud resources.

System Architecture

A layered AWS architecture reflecting the AWS Well-Architected Framework principles.

	Network Layer	VPC with public & private subnets across multiple Availability Zones for fault tolerance.
	Compute Layer	EC2 instances behind an Elastic Load Balancer, scaled via Auto Scaling Groups.
	Storage Layer	Amazon S3 for static assets; Amazon RDS for structured, relational data.
	Security Layer	IAM roles & policies, security groups, and network ACLs enforced at every layer.
	Monitoring Layer	Amazon CloudWatch tracks utilization and triggers threshold-based alerts.

Methodology

A structured, six-phase, week-by-week progression across the 8-week program.

1

Orientation & Onboarding

Introduction to the program, AWS Academy platform, and course structure.

2

Self-Paced Learning

Cloud concepts, core services, security, networking, databases & best practices.

3

Hands-On Labs

Guided labs on the AWS Console within a sandbox environment.

4

Assessments & Quizzes

Periodic module-wise quizzes testing conceptual understanding.

5

Capstone Exercise

Integrating multiple AWS services into a small-scale cloud solution.

6

Evaluation & Certification

Final evaluation across modules, labs & assessments — grade and certificate awarded.

Expected Outcomes



Cloud Fundamentals

A strong conceptual understanding of cloud computing and the AWS ecosystem.



Hands-On Skills

Practical experience provisioning and managing core AWS services.



Architecture Ability

The ability to design basic, secure, and scalable cloud architectures.



Employability

Improved employability through an industry-recognized certification.



Certification Pathway

A foundation for AWS Certified Cloud Practitioner / Solutions Architect.

Certificate of Completion

The Certificate of Virtual Internship, issued by AICTE, the Ministry of Education, the National Internship Portal, and EduSkills, confirms successful completion of the 8-week Cloud Virtual Internship with curriculum provided by AWS Academy.

INSTITUTION

IILM University, Greater Noida

CERTIFICATE ID

4e928ed5d4f6514b118f

STUDENT ID

STU6998634f476331771594575



Grade Awarded

Thank You



Acknowledgement

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References

- AWS Academy Cloud Foundations — aws.amazon.com/training/awsacademy
- Amazon Web Services Documentation — docs.aws.amazon.com
- AICTE Internship Portal — internship.aicte-india.org
- EduSkills Foundation — eduskillsfoundation.org
- AWS Well-Architected Framework — aws.amazon.com/architecture/well-architected